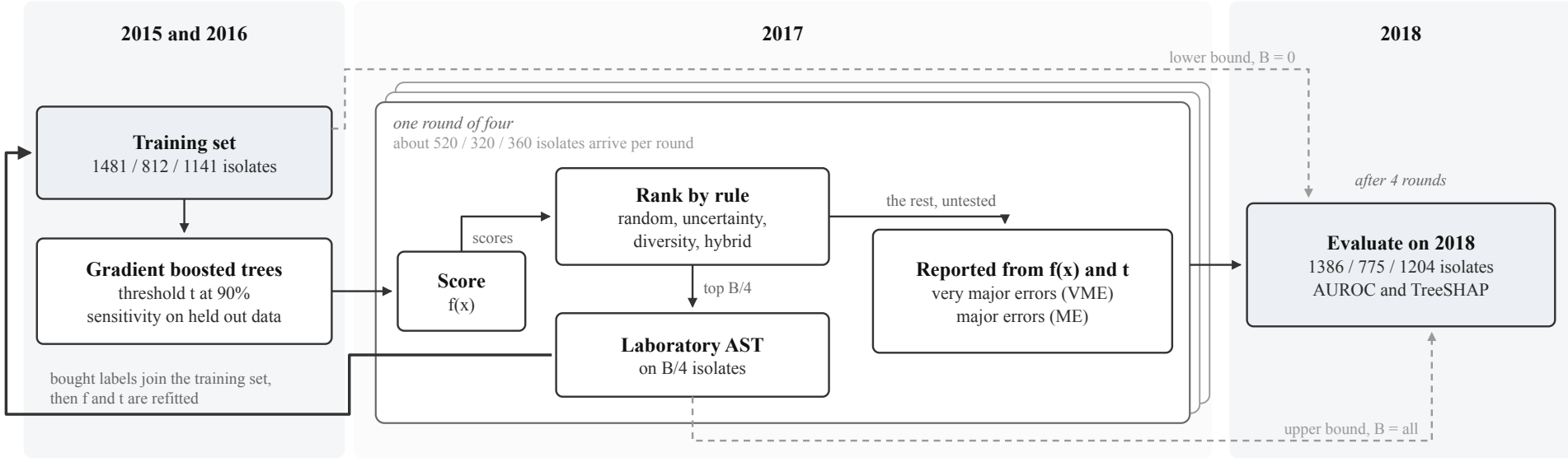


Fig. 1. Study pipeline. A gradient boosted classifier is fitted to the isolates of 2015 and 2016 and a decision threshold t is fixed on held out data at 90% sensitivity. The 3 counts in each box are the isolates available for *E. coli*, *K. pneumoniae* and *S. aureus* at DRIAMS-A. The card drawn in the 2017 band is one of 4 identical rounds, in which the model scores the isolates that arrived, the acquisition rule ranks them, a quarter of the annual budget B is spent on laboratory susceptibility testing (AST), and those labels join the training set before the model and the threshold are refitted. Every isolate that is not bought is reported from the model output alone, and those reports are where the very major and major errors of the pool year accumulate. The dashed arms bypass the loop and carry the 2 bounds the rules are compared against, namely never retraining (B = 0) and testing every arriving isolate (B = all).